



中山大學  
SUN YAT-SEN UNIVERSITY

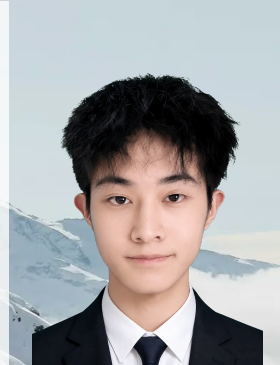
# Yongxue Xu (Jerry)

Undergraduate researcher at Sun Yat-sen University

Video generation • World models • 4D scene understanding

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Google Scholar • LinkedIn



## Profile

Undergraduate researcher working on video understanding, object-centric temporal modeling, and 4D dynamic reasoning. Seeking an industry research internship in multimodal foundation models, video generation, or world models.

Available from October 2026 for 6+ months, through graduation in June 2027.

## Education & Toolkit

### Sun Yat-sen University

SEP 2023 -- JUN 2027

B.Eng. in Intelligent Science and Technology

GPA 3.91 / 4.0 • Major GPA 3.94 / 4.0

Project 985 / 211 / Double First-Class

### RESEARCH

Video understanding/generation, multimodal foundation models, world models, 4D scenes, WAM.

### TOOLS & LANGUAGES

Python, C/C++, MATLAB, PyTorch; Chinese (native), English (CET-6).

## Research Experience

### LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

Co-first author

NeurIPS 2026 submission

Developed [LMM-Track4D](#) for trajectory-grounded multi-turn dialogue: answering spatiotemporal questions about multi-view videos while predicting query-conditioned 3D object trajectories. Combined geometry-aware encoding, persistent state memory, and structured trajectory decoding; co-developed Track4D-Bench.

Substantially outperformed the base model on Track4D-Bench and external benchmarks, especially in dynamics reasoning.

**Contribution:** Led model design and main experiments; co-developed the benchmark.

### Learning to Reason with Persistent Object States for Video Instance Segmentation

Co-first author

ICLR 2027 submission

Developed [POSReasoner](#) for video instance and object segmentation, preserving identities as objects become occluded and reappear. Coupled persistent object memory with verified state transitions to prevent ambiguous observations from corrupting identity history.

Improved long-horizon identity continuity and object recovery with the segmentation models frozen.

**Contribution:** Led method and architecture design, implementation, experiments, and manuscript preparation.

### Graph Domain Adaptation Does Not End with Representation Learning

Co-first author

ICLR 2027 submission

Developed [EviGDA](#) for unsupervised graph domain adaptation: transferring node classifiers from labeled source graphs to unlabeled target graphs by fusing graph-aware and feature-only predictions.

Surpassed existing state-of-the-art methods in overall graph domain adaptation performance.

**Contribution:** Designed the core method and led implementation, evaluation, and manuscript development.

### Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach

First author / TNSE

under review

Developed [DRCIM-ML](#) for robust seed selection in competitive multilayer networks under node-removal attacks, using diffusion-aware role alignment and cross-layer evolutionary search.

Extended RCIM from single-layer to multilayer networks, with a new formulation and a dedicated solver.

**Contribution:** Led problem formulation, method design, experiments, and writing; one patent pending.

## LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue 2026

Chaoyue Li\*, Yongxue Xu\*, Jie Feng\*, and Jiayu Ding<sup>†</sup>

Submitted to *NeurIPS 2026*; [arXiv:2605.19390](#)

## Learning to Reason with Persistent Object States for Video Instance Segmentation 2026

Yongxue Xu\*, Boxue Yang\*, Ziqian Liu, Shaoqiu Zhang, and Linfeng Zhang<sup>†</sup>

Submitted to *ICLR 2027*

## Graph Domain Adaptation Does Not End with Representation Learning 2026

Yongxue Xu\*, Ziqian Liu\*, Enze Zhang, and Maolin Wang<sup>†</sup>

Submitted to *ICLR 2027*

## TRACE: Typed Reference-Aware Conversational Editing for Humanoid Motion Diffusion 2026

Zejian Kang, Yifu Luo, Boyang Shen, Jingxuan Li, Zekai Chen, Shi Zekun, Yongxue Xu, Zhenyu Hou, Bingqian Wu, Shaolei Zhang, Wentao Yang, and Xiangru Huang

Submitted to *ICLR 2027*

## Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach 2026

Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang<sup>†</sup>

Under review at *IEEE Transactions on Network Science and Engineering*

## Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks 2025

Zhonghan Liu, Wenshuo Liu, Yongxue Xu, Ziqian Liu, Yichen Liu, and Shuai Wang<sup>†</sup>

*GBCESC 2025 / Best Paper Award*

\* Equal contribution.    <sup>†</sup> Corresponding author.

## Research Initiatives & Collaborations

Collaborate with researchers at **HKUST**, **SJTU**, and **Westlake University** on video generation and multimodal spatiotemporal understanding; explore AI-assisted research with peers in **InkMind.AI**.

**TRACE (humanoid motion)**: reproduced baseline and comparison methods, adapted evaluators, and converted motion demos for Unitree G1.

**Urban road-network analysis** (GBCESC 2025, Best Paper): contributed dataset construction, GNN/DQN baseline reproduction, and experimental visualization for critical-node identification.

## Research Tools

**AstraDraw**. Developed a reference-guided workflow for editable PowerPoint research figures, reusable visual assets, and iterative local refinement. Used in my own papers; attracted [100+ GitHub stars](#).

**CCFA-Skills**. Co-developed with **InkMind.AI collaborators**: a coordinated toolkit of 17 research skills spanning literature, experiments, writing, figures, review, and submission. Integrated AstraDraw for reference-guided figure authoring.

## Awards & Competitions

Sun Yat-sen University First-Class Scholarship 2023–2026

Chinese Mathematics Competition (CMC), Guangdong Region – Second Prize Dec. 2024

Mathematical Contest in Modeling (MCM), USA – Honorable Mention Feb. 2025

National Undergraduate Smart Car Competition, South China Region – Second Prize Aug. 2025

**Beyond Research**: Table tennis, music, and nature travel.